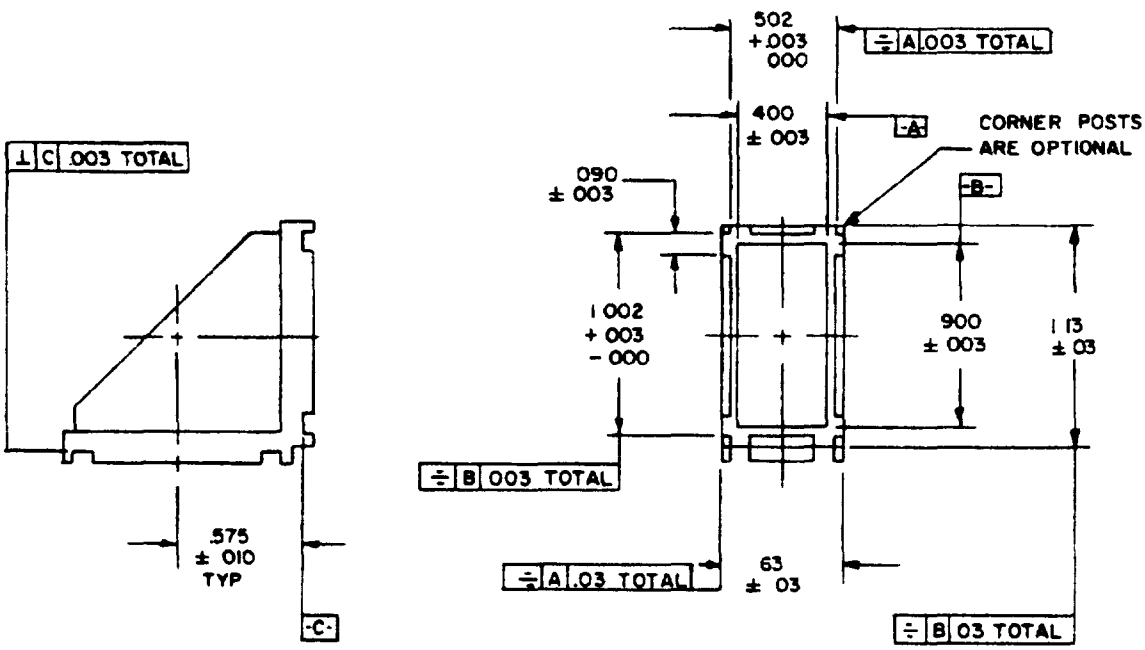
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLIES, RIGID, CAST, 90-DEGREE H-PLANE BEND, MITER AND RADIUS-BACK MITER, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assemblies described herein shall consist of this document and the latest issue of Specification MIL-W-3970.



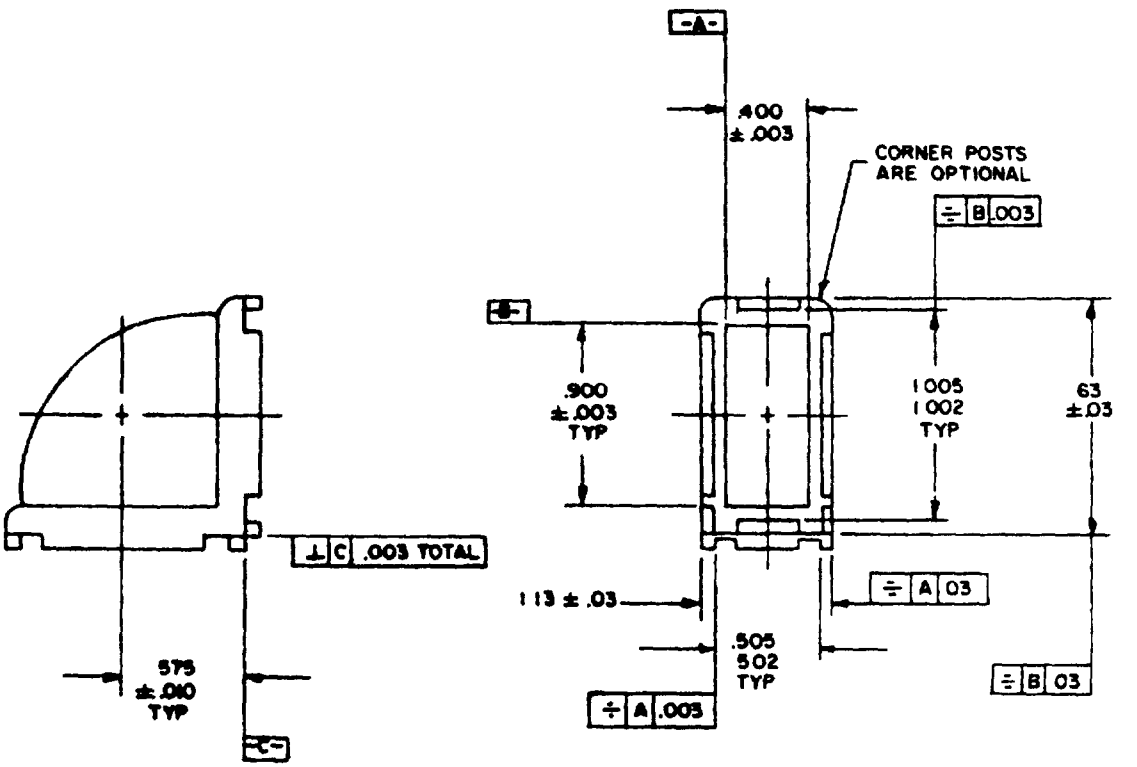
INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.400	10.16	.900	22.86
.010	.25	.502	12.75	1.002	25.45
.03	.8	.575	14.60	1.13	28.7
.090	2 29	.63	16 0		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 1. Dimensions for miter and full corral configuration, for dash number 001

FSC 5985

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RADIUS-BACK MITER

INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.502	12.75	.900	22.86
.010	.25	.505	12.83	1.002	25 45
.03	.8	.575	14.60	1.005	25.53
.400	10.16	.63	16.0	1.13	28.7

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.

FIGURE 2. Dimensions for radius-back miter and full corral configuration, for dash number 002.

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REQUIREMENTS

Physical characteristics:
Dimensions and configuration: See figures 1 and 2.
Operating temperature range: -55° to +200°C.
Material: D712.0, temper F, aluminum alloy per QQ-A-601 or C712.1 per QQ-A-371.
Surfaces (sandblasting after machining permissible):
Internal: $63\sqrt{}$ microinches per ANSI B46.1.

External: $125\sqrt{}$ microinches per ANSI B46.1.

Fungus resistance: Materials nutrient to fungus shall not be used.

Pressurization: 45 lb_f/in^2 gage maximum.

Penetrant inspection. No indication of any cracks at any location when tested per MIL-I-6866, type I, method B.

Electrical characteristics:
Frequency range: 8.2 to 12.4 GHz.
VSWR 1.04:1 maximum.
Power.
Peak - 250 kW.
Average - 10 kW.

Part number M3970/29-(dash number from figures 1 and 2).

NOTE The mating waveguides are M85/1-077, -078, and -178.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE MIL-W-3970/39

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED S/N 0102-014-1802

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